Biases & Fallacies — Complete Reference

PART I — BIASES

1. Belief & Decision Biases

Bias	Brief explanation
Confirmation bias	Seeking and favoring evidence that supports what you already believe
Myside bias	Generating arguments only for your own position
Belief perseverance	Holding a belief after its evidence is disproven
Backfire effect	Contrary evidence strengthens the original belief
Congruence bias	Testing only your own hypothesis, not alternatives
Semmelweis reflex	Reflexively rejecting information that contradicts a paradigm
Anchoring	Over-weighting the first number or fact encountered
Adjustment bias	Insufficiently adjusting away from the anchor
Framing effect	Same facts produce different choices depending on wording
Loss aversion	Losses feel roughly twice as painful as equivalent gains
Endowment effect	Valuing something more simply because you own it
Status quo bias	Default preference for leaving things unchanged
System justification	Defending existing arrangements as fair and legitimate
Sunk cost fallacy	Continuing because of unrecoverable past investment
Escalation of commitment	Increasing investment in a visibly failing course
Choice-supportive bias	Remembering chosen options more favorably than they were
Post-purchase rationalization	Retroactively justifying a poor purchase
Decoy effect	A third inferior option shifts preference between two others
Ambiguity aversion	Preferring known risks over unknown probabilities
Zero-risk bias	Preferring elimination of a small risk over larger overall reduction
Certainty effect	Overvaluing outcomes that are guaranteed
Pseudocertainty effect	Risk-averse on likely gains, risk-seeking on likely losses

Bias	Brief explanation
Omission bias	Judging harmful action as worse than equally harmful inaction
Action bias	Preferring to act even when waiting is better
Default effect	Accepting whatever option is pre-selected
Analysis paralysis	Over-deliberation prevents any decision
Information bias	Seeking data that cannot change the decision
Less-is-better effect	Preferring the smaller option when items are judged separately
Distinction bias	Differences look larger when options are compared side by side
Hyperbolic discounting	Strongly preferring smaller immediate rewards
Present bias	Undervaluing future costs and benefits
Projection bias	Assuming future preferences will match current ones
Restraint bias	Overestimating your ability to resist temptation

2. Probability & Estimation Biases

Bias	Brief explanation
Availability heuristic	Judging likelihood by how easily examples come to mind
Representativeness heuristic	Judging by resemblance to a stereotype, ignoring base rates
Base rate neglect	Ignoring background frequency when reading evidence
Conjunction fallacy	Rating a specific combination as more likely than one of its parts
Gambler's fallacy	Believing independent events "even out"
Hot-hand fallacy	Believing a streak predicts continued success
Clustering illusion	Seeing patterns in random data
Illusory correlation	Perceiving a relationship where none exists
Insensitivity to sample size	Treating small samples as reliable
Law of small numbers	Generalizing confidently from few cases
Neglect of probability	Ignoring likelihood entirely under uncertainty
Subadditivity effect	Whole judged less likely than the sum of its parts
Denomination effect	Spending differs by bill size for identical amounts
Money illusion	Confusing nominal and real value
Mental accounting	Treating identical money differently by mental category
Anchoring on round numbers	Estimates cluster on salient figures
Scope insensitivity	Willingness to pay barely rises with magnitude of benefit
Compassion fade	Concern decreases as the number of victims grows
Identifiable victim effect	One named person moves you more than statistics

3. Self & Confidence Biases

Bias	Brief explanation
Overconfidence effect	Confidence exceeds accuracy
Dunning–Kruger effect	Low competence pairs with inflated self-assessment
Illusory superiority	Rating yourself above average on most traits
Illusion of control	Overestimating influence over outcomes
Optimism bias	Underestimating your own risk of bad outcomes
Pessimism bias	Overestimating likelihood of negative outcomes
Planning fallacy	Systematically underestimating time, cost, and risk
Self-serving bias	Crediting success internally, blaming failure externally
Egocentric bias	Over-weighting your own contribution and perspective
Illusion of transparency	Assuming your feelings are obvious to others
Spotlight effect	Overestimating how much others notice you
Curse of knowledge	Assuming others share your background knowledge
False consensus effect	Assuming others agree with you more than they do
Bias blind spot	Recognizing bias in others but not yourself
Naive realism	Believing you see reality objectively
Naive cynicism	Assuming others are more self-interested than they are
Hard–easy effect	Overconfident on hard tasks, underconfident on easy ones
Impostor phenomenon	Persistent doubt about deserved competence
Overclaiming	Claiming familiarity with things that don't exist

4. Memory Biases

Bias	Brief explanation
Hindsight bias	Past outcomes seem more predictable than they were
Rosy retrospection	Remembering the past more fondly than experienced
Fading affect bias	Negative emotions fade faster than positive ones
Peak-end rule	Judging an experience by its peak and ending only
Duration neglect	Length of an experience barely affects the memory
Recency effect	Best recall for the most recent items
Primacy effect	Best recall for the first items
Serial position effect	Middle items are recalled worst
Misinformation effect	Later information corrupts an earlier memory
Source confusion	Remembering content but misattributing its source
Cryptomnesia	Mistaking a recalled memory for an original idea
False memory	Confident recall of events that never occurred
Consistency bias	Misremembering past attitudes as matching current ones
Telescoping effect	Misplacing when remembered events occurred
Suggestibility	Absorbing others' suggestions into your own memory
Levelling and sharpening	Losing details while exaggerating others in retelling
Zeigarnik effect	Unfinished tasks are remembered better
Von Restorff effect	Distinctive items stand out in memory
Google effect	Poorer recall for information you know is retrievable

5. Social & Interpersonal Biases

Bias	Brief explanation
Fundamental attribution error	Explaining others' behavior by character, yours by circumstance
Actor–observer bias	Asymmetry in explaining your actions versus others'
Ultimate attribution error	Attributing out-group failings to inherent traits
Halo effect	One good trait inflates judgment of unrelated traits
Horn effect	One bad trait taints overall judgment
Physical attractiveness stereotype	Assuming attractive people have better qualities
In-group favoritism	Preferential treatment for your own group
Out-group homogeneity	Seeing outsiders as more alike than they are
Cross-race effect	Poorer recognition of faces from other groups
Affinity bias	Preferring people similar to yourself
Similar-to-me effect	Rating similar candidates higher in evaluation
Stereotyping	Applying group generalizations to individuals
Implicit bias	Unconscious associations affecting behavior
Just-world hypothesis	Believing people get what they deserve
Victim blaming	Attributing harm to the victim's choices
Bandwagon effect	Adopting beliefs because they are popular
Conformity bias	Aligning judgment with the group
Groupthink	Cohesion suppresses dissent and critical review
Group attribution error	Treating one member's view as the group's
Authority bias	Over-weighting opinions of authority figures
Halo of expertise	Assuming competence transfers across domains
Social desirability bias	Reporting what looks socially acceptable
Reactance	Resisting a request because it constrains freedom

Bias	Brief explanation
Reciprocity bias	Feeling obliged to return favors disproportionately
Mere exposure effect	Liking things more with repeated exposure
Not-invented-here syndrome	Devaluing ideas from outside your group
IKEA effect	Overvaluing what you helped build
Pygmalion effect	Expectations cause the expected performance
Golem effect	Low expectations depress performance
Bystander effect	Diffused responsibility reduces intervention
Moral licensing	A good act licenses a later bad one
Moral luck	Judging identical intent differently by outcome
Trait ascription bias	Seeing yourself as variable, others as predictable
Illusion of asymmetric insight	Believing you know others better than they know you

6. Perception & Attention Biases

Bias	Brief explanation
Attentional bias	Recurrent thoughts skew what you notice
Salience bias	Prominent features dominate judgment
Frequency illusion	A newly noticed thing seems suddenly everywhere
Selective perception	Expectations filter what is perceived
Contrast effect	Judgment shifts based on what preceded it
Anchoring by exposure	Prior stimuli set the comparison standard
Pareidolia	Seeing meaningful patterns in noise
Apophenia	Perceiving connections among unrelated things
Focusing illusion	Over-weighting one aspect when predicting happiness
Empathy gap	Underestimating the force of emotional states
Hot–cold empathy gap	Misjudging your own behavior in a different emotional state
Illusion of validity	Overconfidence in interpretations of limited data
Illusory truth effect	Repetition makes a statement feel true
Processing fluency	Easily processed information feels more true
Rhyme-as-reason effect	Rhyming statements are judged more accurate
Picture superiority effect	Images are retained better than words
Ambiguity effect	Avoiding options with unknown probabilities
Semantic satiation	Repetition drains a word of meaning
Automation bias	Over-trusting automated output over your own judgment
Algorithm aversion	Distrusting algorithmic advice despite better accuracy

7. Research, Data & Methodological Biases

Bias	Brief explanation
Selection bias	Sample systematically unrepresentative of the population
Sampling bias	Non-random selection distorts results
Convenience sampling bias	Using whoever is easiest to reach
Self-selection bias	Volunteers differ from non-volunteers
Survivorship bias	Analyzing only the cases that persisted
Attrition bias	Dropouts differ systematically from completers
Non-response bias	Non-responders differ from responders
Berkson's paradox	Hospital-based sampling creates false associations
Immortal time bias	Misclassified follow-up time inflates benefit
Length-time bias	Screening over-detects slow-progressing cases
Lead-time bias	Earlier detection lengthens apparent survival only
Prevalence–incidence bias	Studying survivors of a condition, not new cases
Measurement bias	Faulty instrument or procedure distorts data
Instrument decay	Measurement quality drifts over time
Recall bias	Differential accuracy of remembered exposure
Reporting bias	Selective disclosure of information
Response bias	Answers shaped by question design or context
Acquiescence bias	Tendency to agree regardless of content
Extreme / central-tendency response bias	Systematic preference for scale endpoints or midpoints
Order effects	Item sequence changes responses
Question-wording bias	Loaded phrasing steers answers
Interviewer bias	Interviewer's manner influences answers
Observer / experimenter bias	Researcher expectations shape observation

Bias	Brief explanation
Observer-expectancy effect	Subtle cues elicit the expected result
Hawthorne effect	Behavior changes because it is being observed
Demand characteristics	Participants infer and enact the hypothesis
Placebo effect	Improvement from expectation, not treatment
Confounding bias	An uncontrolled third variable distorts the link
Collider bias	Conditioning on a common effect creates spurious links
Detection bias	Outcome sought harder in one group
Performance bias	Groups receive different co-interventions
Publication bias	Positive, novel results are published more
Time-lag bias	Negative results published later
Language bias	Only same-language studies included
Citation bias	Supportive studies cited more heavily
Funding / sponsorship bias	Results skew toward the funder's interest
Outcome reporting bias	Only favorable outcomes reported
p-hacking	Analytic choices tuned until significance appears
HARKing	Hypothesizing after results are known
Multiple comparisons problem	Many tests guarantee false positives
Overfitting	Model captures noise as signal
Data dredging	Searching data for any reportable pattern
Look-elsewhere effect	Significance found somewhere in a broad search
Regression to the mean	Extremes drift toward average without cause
Ecological fallacy	Group-level findings applied to individuals
Atomistic fallacy	Individual-level findings applied to groups
Aggregation bias	Combining groups obscures or reverses effects

8. Institutional, Organizational & Economic Biases

Bias	Brief explanation
Institutional bias	Policies and practices systematically favor some groups
Structural bias	Bias embedded in systems, not individual intent
Path dependence	Past choices constrain present options
Bureaucratic inertia	Processes persist beyond their usefulness
Optimism in forecasting	Institutional plans systematically understate cost
Strategic misrepresentation	Deliberate underbidding to win approval
Agency bias	Agents' incentives diverge from principals'
Short-termism	Near-term metrics crowd out long-term value
Metric fixation	Managing to measures rather than outcomes
Goodhart's law effect	A measure loses meaning once it becomes a target
Home bias	Over-investing in domestic or familiar assets
Familiarity bias	Preferring known companies or instruments
Disposition effect	Selling winners early, holding losers too long
Recency bias in markets	Extrapolating recent performance forward
Herding	Following the crowd into positions
Narrative fallacy	Fitting a tidy story onto random outcomes
Hindsight in investing	Believing past moves were obvious
Attribution bias in performance	Skill claimed for luck-driven results

9. Machine Learning & AI Biases

Bias	Brief explanation
Historical bias	Past inequities encoded in training data
Representation bias	Subgroups underrepresented in the dataset
Sampling bias (ML)	Training distribution differs from deployment
Label bias	Subjective, noisy, or inconsistent annotation
Proxy bias	A feature stands in for a protected attribute
Aggregation bias (ML)	One model applied to heterogeneous subgroups
Evaluation bias	Benchmarks unrepresentative of real use
Deployment bias	System used outside its intended context
Feedback loop bias	Predictions shape data that retrains the model
Popularity bias	Recommenders amplify already-popular items
Position bias	Ranking position drives clicks, not relevance
Exposure bias	Model only learns from what was shown
Automation bias (ML)	Human defers to model output uncritically
Anchoring on model output	Model suggestion sets the human's baseline
Emergent / amplification bias	Model magnifies small disparities in data
Simpson's paradox in models	Subgroup effects reverse in pooled data

PART II — FALLACIES

10. Formal Fallacies (invalid structure)

Fallacy	Brief explanation
Affirming the consequent	If A→B; B is true, so A must be true
Denying the antecedent	If A→B; A is false, so B must be false
Affirming a disjunct	A or B; A is true, so B is false
Denying a conjunct	Not both A and B; not A, so B
Undistributed middle	All A are C; all B are C; therefore A is B
Illicit major / minor	A term is broader in the conclusion than the premise
Exclusive premises	Two negative premises yield no valid conclusion
Existential fallacy	Concluding something exists from universal premises
Masked-man fallacy	Substituting terms not known to be identical
Quantifier shift	Swapping "every … some" order illegitimately
Modal fallacy	Confusing possibility with necessity
Fallacy of the four terms	A syllogism using four distinct terms
Base rate fallacy	Evidence read without prior probability
Conjunction fallacy	Compound claim rated more probable than a conjunct
Inverse fallacy	Confusing P(A|B) with P(B|A)

11. Fallacies of Relevance

Fallacy	Brief explanation
Ad hominem (abusive)	Attacking the arguer's character
Ad hominem (circumstantial)	Dismissing an argument by the arguer's interests
Tu quoque	"You do it too" offered as refutation
Poisoning the well	Discrediting a source before it speaks
Guilt by association	Rejecting a claim by who else holds it
Genetic fallacy	Judging a claim solely by its origin
Straw man	Attacking a distorted version of the position
Steel-man reversal / hollow man	Inventing a weak opponent nobody holds
Red herring	Introducing an irrelevant distraction
Appeal to authority	True because an authority asserts it
Appeal to false authority	Citing an authority outside their field
Appeal to popularity	True because most people believe it
Appeal to tradition	True because it has always been done
Appeal to novelty	True because it is new
Appeal to nature	Natural is therefore good or correct
Appeal to emotion	Feeling substituted for evidence
Appeal to fear	Threat used in place of argument
Appeal to pity	Sympathy used in place of argument
Appeal to flattery	Praise used to win assent
Appeal to spite	Resentment used to win assent
Appeal to ridicule	Mockery presented as refutation
Appeal to consequences	A claim is false because outcomes are unwelcome
Wishful thinking	Believing because it is desirable
Appeal to ignorance	Unproven false therefore true, or vice versa

Fallacy	Brief explanation
Argument from incredulity	"I can't imagine it, so it's false"
Appeal to force	Coercion replacing reasoning
Moralistic fallacy	Inferring facts from what ought to be
Naturalistic fallacy	Inferring values from what is
Two wrongs make a right	Justifying wrong by citing another wrong
Bulverism	Explaining why someone is wrong instead of showing it

12. Fallacies of Ambiguity

Fallacy	Brief explanation
Equivocation	A key word shifts meaning mid-argument
Amphiboly	Ambiguous grammar yields a false conclusion
Accent / emphasis	Stress or omission alters meaning
Composition	A part's property assumed true of the whole
Division	The whole's property assumed true of each part
Reification	Treating an abstraction as a concrete thing
Definist fallacy	Defining a term to win the argument
No true Scotsman	Redefining a category to exclude counterexamples
Motte-and-bailey	Retreating to a defensible claim after asserting a strong one
Continuum fallacy	Denying a distinction because boundaries are fuzzy
Ecological equivocation	Sliding between group and individual referents
Suppressed correlative	Redefining a term so its opposite is impossible

13. Fallacies of Presumption

Fallacy	Brief explanation
Begging the question	The conclusion is assumed in the premises
Circular reasoning	A supports B and B supports A
Loaded question	The question embeds an unproven assumption
Complex question	Two questions fused so one answer covers both
False dilemma	Only two options presented when more exist
False equivalence	Treating unlike things as comparable
False analogy	Analogy breaks at the relevant point
Middle ground	Compromise assumed to be correct
Nirvana fallacy	Rejecting a solution for not being perfect
Perfect solution fallacy	Demanding total success as the standard
Special pleading	Exempting a case without principled reason
Moving the goalposts	Raising the standard once it is met
Ad hoc rescue	Inventing an excuse to save a refuted claim
Unfalsifiability	Framing a claim so nothing could disprove it
Shifting the burden of proof	Demanding the other side disprove your claim
Argument from silence	Absence of mention treated as evidence
Fallacy of the single cause	Assuming one cause for a multi-causal event
Historian's fallacy	Judging past actors by information they lacked
Presentism	Applying current standards to past contexts
Chronological snobbery	Dismissing older ideas as necessarily inferior
Just-world fallacy	Assuming outcomes reflect desert
Fallacy fallacy	Concluding a claim is false because its argument was bad

14. Fallacies of Weak Induction & Causation

Fallacy	Brief explanation
Hasty generalization	Conclusion drawn from too few cases
Anecdotal evidence	A vivid case overrides systematic data
Cherry picking	Selecting only supportive evidence
Texas sharpshooter	Drawing the boundary around a random cluster
Overgeneralization	Applying a valid narrow finding too broadly
Slippery slope	Small step assumed to cause an extreme end
Post hoc ergo propter hoc	Sequence mistaken for causation
Cum hoc ergo propter hoc	Correlation mistaken for causation
Reverse causation	Cause and effect direction inverted
Third-cause fallacy	Ignoring a common cause of both variables
Regression fallacy	Natural variation credited to an intervention
Gambler's fallacy	Independent events assumed to self-correct
Inverse gambler's fallacy	Rare outcome taken as evidence of many trials
Hot-hand fallacy	Streak assumed to predict continuation
Broken window fallacy	Counting visible gains, ignoring unseen costs
Sunk cost fallacy	Past cost used to justify future spending
McNamara fallacy	Only measurable factors treated as real
Simpson's paradox	Pooled trend contradicts every subgroup
Survivorship reasoning	Conclusions drawn from survivors only
Prosecutor's fallacy	Match probability confused with guilt probability
Defense attorney's fallacy	Dismissing evidence as individually weak
Misleading vividness	Dramatic instance outweighs the statistics
False precision	Spurious exactness implies reliability
Slothful induction	Denying the conclusion the evidence supports

Fallacy	Brief explanation
Proof by verbosity	Overwhelming volume presented as argument
Argument by repetition	Repeating a claim in place of supporting it
Argument by gibberish	Impressive-sounding nonsense as support
Thought-terminating cliché	A stock phrase ends inquiry
Kettle logic	Multiple mutually inconsistent defenses
Whataboutism	Deflecting criticism onto another target
Gish gallop	Rapid-fire weak claims outpacing rebuttal

Quick Distinction

	Bias	Fallacy
Nature	Systematic error in thinking or measurement	Error in the structure of an argument
Location	In the mind, method, or data	In the reasoning as stated
Awareness	Usually unconscious	Can be accidental or deliberate
Fix	Better process, blinding, checklists	Better logic, explicit premises